

Quiz: April 5, 2023
Multivariate Meta-Analysis
STAT490 [Spring 2023]

Max score: 20 (4 questions x 5)

Effect size: SMD

	pubyear	npat	PD	AL	var_PD	cov_PD_AL	var_AL
Pihlstrom	1983	14	0.47	-0.32	0.0075	0.003	0.0077
Lindhe	1982	15	0.2	-0.6	0.0057	0.0009	0.0008
Knowles	1979	78	0.4	-0.12	0.0021	0.0007	0.0014
Ramfjord	1987	89	0.26	-0.31	0.0029	0.0009	0.0015
Becker	1988	16	0.56	-0.39	0.0148	0.0072	0.0304

Question 1: Perform Univariate fixed-effects meta-analysis using PD and AL separately.

```
Call: mvmeta(formula = PD ~ 1, S = berkey98[, "var_PD"], data = berkey98,
method = "fixed")
```

```
Univariate fixed-effects meta-analysis
Dimension: 1
```

```
Fixed-effects coefficients
      Estimate Std. Error      z Pr(>|z|) 95%ci.lb 95%ci.ub
(Intercept)  0.3472    0.0289 12.0141  0.0000  0.2906  0.4038 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Univariate Cochran Q-test for heterogeneity:
Q = 12.8213 (df = 4), p-value = 0.0122
I-square statistic = 68.8%
```

```
5 studies, 5 observations, 1 fixed and 0 random-effects parameters
logLik      AIC      BIC
2.1357 -2.2715 -2.6620
```

```
Call: mvmeta(formula = AL ~ 1, S = berkey98[, "var_AL"], data = berkey98,
method = "fixed")
```

```
Univariate fixed-effects meta-analysis
Dimension: 1
```

```
Fixed-effects coefficients
      Estimate Std. Error      z Pr(>|z|) 95%ci.lb 95%ci.ub
(Intercept) -0.3926    0.0189 -20.7503  0.0000 -0.4297 -0.3555 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Univariate Cochran Q-test for heterogeneity:
Q = 112.0808 (df = 4), p-value = 0.0000
I-square statistic = 96.4%
```

```
5 studies, 5 observations, 1 fixed and 0 random-effects parameters
logLik      AIC      BIC
-46.3529  94.7059  94.3153
```

Question 2: Compare your results above with the following result of *fixed-effect multivariate meta-analysis*.

Multivariate fixed-effects meta-analysis

Dimension: 2

Fixed-effects coefficients

	Estimate	Std. Error	z	Pr(> z)	95%ci.lb	95%ci.ub	
PD	0.3072	0.0286	10.7513	0.0000	0.2512	0.3632	***
AL	-0.3944	0.0186	-21.1471	0.0000	-0.4309	-0.3578	***

Multivariate Cochran Q-test for heterogeneity:

Q = 128.2267 (df = 8), p-value = 0.0000

I-square statistic = 93.8%

The results are consistent between the two, though the test statistics and confidence intervals around each effect size differ slightly. Testing the effect sizes independently shows that both are significantly different to 0. The multivariate meta-analysis produces similar results when taking the relationship between the effect sizes into account. However, the test for heterogeneity resulted in an extremely small p-value, suggesting there is significant between-study variability. Therefore, the results from the univariate fixed-effect models and multivariate fixed-effect model may not be as reliable as a random-effects model.

Question 3: Perform Univariate random-effects meta-analysis using PD and AL separately. Use REML method for estimating τ

```
Call: mvmeta(formula = PD ~ 1, S = berkey98[, "var_PD"], data = berkey98,
  method = "reml")
```

```
Univariate random-effects meta-analysis
Dimension: 1
Estimation method: REML
```

```
Fixed-effects coefficients
      Estimate Std. Error      z Pr(>|z|) 95%ci.lb 95%ci.ub
(Intercept)  0.3606    0.0592  6.0906  0.0000  0.2445  0.4766 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Between-study random-effects (co)variance components
Std. Dev
0.1090
```

```
Univariate Cochran Q-test for heterogeneity:
Q = 12.8213 (df = 4), p-value = 0.0122
I-square statistic = 68.8%
```

```
5 studies, 5 observations, 1 fixed and 1 random-effects parameters
logLik    AIC    BIC
1.3588  1.2823  0.0549
```

```
Univariate random-effects meta-analysis
Dimension: 1
Estimation method: REML
```

```
Fixed-effects coefficients
      Estimate Std. Error      z Pr(>|z|) 95%ci.lb 95%ci.ub
(Intercept) -0.3457    0.0885 -3.9058  0.0001 -0.5191 -0.1722 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Between-study random-effects (co)variance components
Std. Dev
0.1819
```

```
Univariate Cochran Q-test for heterogeneity:
Q = 112.0808 (df = 4), p-value = 0.0000
I-square statistic = 96.4%
```

```
5 studies, 5 observations, 1 fixed and 1 random-effects parameters
logLik    AIC    BIC
0.2010  3.5980  2.3706
```

Question 4: Compare your results above with the following result of *random-effects multivariate meta-analysis*.

Multivariate random-effects meta-analysis

Dimension: 2

Estimation method: REML

	Estimate	Std. Error	z	Pr(> z)	95%ci.lb	95%ci.ub	
PD	0.3534	0.0588	6.0057	0.0000	0.2381	0.4688	***
AL	-0.3392	0.0879	-3.8589	0.0001	-0.5115	-0.1669	***

Between-study random-effects (co)variance components Structure: General positive-definite

	Std. Dev	Corr
PD	0.1083	PD
AL	0.1807	0.6088

Multivariate Cochran Q-test for heterogeneity:

Q = 128.2267 (df = 8), p-value = 0.0000

I-square statistic = 93.8%

The results are consistent once again. Each of the univariate random-effects models showed that the respective effect size was significantly different to 0, and this is supported by the results of the multivariate meta-analysis. The only differences between the two approaches are the parameter estimates and their corresponding standard error, and thereby the corresponding confidence intervals. This difference is a result of the correlation *between* the two effect sizes, which is not accounted for in the univariate meta-analyses.

It should be noted that the p-values were larger for the random-effects models, which is expected because there are now two sources of variability for each effect size: within-study and between-study. This also leads to wider confidence intervals